

CISCO VIRTUAL INTERNSHIP 2026

INDIVIDUAL PROJECT CONTRIBUTION SUMMARY

Candidate Name	Nikhil Singh Chauhan
AICTE Registration ID	STU666b367584a5f1718302325
College Name	Indian Institute of Technology, Patna
Technology Track	Cyber Security & Enterprise Networking
Project Title	Secure Hybrid Data Center Network Architecture & Multi-Cloud Workload Protection

1. Individual Contribution Breakdown

- Hybrid IPsec VPN & Edge Firewall: Configured Site-to-Site IKEv2 IPsec VPN (AES-256-GCM / SHA-256 / DH 14) and Identity NAT on Cisco ASA 5506-X to seamlessly connect on-premise OpenShift clusters to AWS Transit Gateway.
- Cloud VPC Segmentation & Security Groups: Authored Terraform code for 3-tier subnets and SG-to-SG referencing rules, ensuring complete isolation of database tiers from public ingress.
- Kubernetes Microservice Security: Created Calico default-deny NetworkPolicies, Restricted Pod Security Standards, and Istio STRICT mTLS 1.3 manifests.
- Zero-Trust Remote Faculty Access: Designed Cisco AnyConnect SSL VPN profiles with Cisco Duo MFA push notifications and Cisco ISE posture compliance assessment.
- DevSecOps Pipeline Automation: Integrated Semgrep SAST, Checkov IaC scanning, and Trivy container scanning into automated CI/CD security gates.
- Packet Tracer Lab Simulation: Built and validated the entire multi-device network topology with complete CLI configuration scripts.

2. Cisco Packet Tracer Deliverables

The submission includes tested configuration files for Enterprise Core Router, ASA Perimeter Firewall, Data Center Switch, ISP Router, Cloud Transit Gateway Router, and Cloud Workload Switch.

Declaration: I hereby declare that this project work is original and completed in accordance with Cisco Networking Academy and AICTE guidelines.

Candidate Signature: _____ *Nikhil* _____

Date: 30th August 2026